

**Cool summers are stormy summers: precession reorganizes extratropical
cyclone activity through the quiet season**

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7 ABSTRACT: Extratropical cyclones peak in winter, while orbital precession reshapes the sea-
8 sonal insolation cycle most strongly in summer. Thus storm tracks might be expected to ignore the
9 precession cycle. This is examined with twelve equilibrium simulations using a coupled climate
10 model AWI-ESM2, spanning a full precession cycle at 30° longitude increments under fixed eccen-
11 tricity and obliquity. Extratropical cyclones are explicitly tracked based on 6-hourly atmospheric
12 pressure fields for both hemispheres. While the annual-mean cyclone response to precession is
13 muted, substantial seasonal variability emerges. In the North Atlantic, the North Pacific, and the
14 Southern Ocean the response concentrates in local summer, where storm activity ranges across
15 the phases by up to 68% of its climatological mean, while winter changes stay below about 10%.
16 We also find summers spent near perihelion feature suppressed storminess, whereas summers near
17 aphelion experience intensified cyclone activity, yielding antiphase midlatitude storm variability
18 between the two hemispheres over the precession cycle. This signal is carried by cyclone number
19 and position, and it follows the pronounced summer response of the Eady growth rate driven
20 primarily by vertical wind shear in all three basins. Monthly storm responses are phase-locked to
21 the calendar timing of perihelion, lagged by 1-2 months due to surface thermal inertia. Precession
22 therefore acts on extratropical storm tracks through their quiet season. Summer-sensitive proxy
23 archives should record a strong and hemispherically antiphased storminess signal at the precession
24 period that annually integrating archives would miss.

25 SIGNIFICANCE STATEMENT: Long orbital cycles drive global climate variability, and the
26 strongest of these, precession, reshapes sunlight mainly in summer, while mid-latitude storms
27 happen mainly in winter. This creates an apparent disconnect between precessional forcing and
28 storm track variability. Climate model simulations covering a full precession cycle show that
29 the orbit nevertheless controls storminess by acting through summer. When perihelion falls in a
30 hemisphere's summer, that summer has fewer mid-latitude storms and the storm belt sits farther
31 poleward, and the two hemispheres evolve in antiphase through the precession cycle. This tells
32 researchers where to look for orbital storm signals in geological records, namely in archives
33 sensitive to summer, not to the yearly average.

34 1. Introduction

35 Extratropical cyclones govern midlatitude weather systems. They transport the majority of pole-
36 ward heat and moisture outside the tropics, supply a substantial fraction of midlatitude precipitation,
37 and aggregate into the canonical storm-track pathways of the North Atlantic, North Pacific, and
38 Southern Ocean (Hoskins and Valdes 1990; Chang et al. 2002; Hoskins and Hodges 2002, 2005).
39 These cyclones develop via baroclinic instability driven by the meridional temperature gradient
40 (Eady 1949), a gradient that reaches its maximum steepness during winter. Cyclone activities con-
41 sequently peak in the cold season, with consistent wintertime maxima (e.g., frequency, strength)
42 evident in both the Northern and Southern Hemispheres (Simmonds and Keay 2000). Midlatitude
43 cyclogenesis is therefore most favorable during winter.

44 Precession, by contrast, exerts its primary forcing on warm seasons. Slow variations in the
45 longitude of perihelion represent the leading orbital control on seasonal insolation cycles. This
46 mechanism redistributes incoming solar radiation, and the amplitude of insolation anomalies
47 scales proportionally with a season's baseline solar irradiance (Berger 1978; Laskar et al. 2004).
48 Midlatitude summertime insolation fluctuates by over 100 W m^{-2} across a full precession cycle,
49 whereas midwinter insolation experiences only minor orbital modulation. This stark contrast
50 explains why orbital climate frameworks single out summertime solar forcing as the most impactful
51 component of climate changes (Huybers 2006). Since extratropical cyclone activity is inherently
52 winter-favored, this invites a simple expectation that precession should exert negligible control
53 over midlatitude storm tracks. But this expectation has never been put to a clean test.

54 Previous studies have looked closely at precessional signals, and they find seasonal shifts in
55 climate. This is well established in monsoon research, from the classic orbital experiments of
56 Kutzbach and Guetter (1986) to single-forcing designs that isolate precession explicitly (Bosmans
57 et al. 2015; Merlis et al. 2013), and cave records show the northern and southern monsoons
58 swinging in antiphase as perihelion migrates through the year (Wang et al. 2008). The extratropics
59 have received far less attention. Kaspar et al. (2007) simulated a stronger and poleward-shifted
60 North Atlantic winter storm track for the Eemian from two orbital time slices. Brayshaw et al.
61 (2010) linked Holocene orbital shifts to North Atlantic and Mediterranean winter storm tracks
62 via changes in baroclinicity. And Varma et al. (2012) found the Southern Hemisphere westerlies
63 migrating under Holocene orbital forcing. Explicit cyclone tracking has entered paleoclimate
64 work mainly at the Last Glacial Maximum, where ice sheets dominate the forcing (Pinto and
65 Ludwig 2020; Raible et al. 2021), and in transient simulations of the last millennium (Hess and
66 Chemke 2025). For tropical cyclones the precessional response has recently been isolated in high-
67 resolution paleoclimate simulations (Raavi et al. 2023). To our knowledge, no previous study has
68 applied explicit cyclone tracking across a full precession cycle to isolate the orbital response of the
69 extratropical storm tracks.

70 This missing piece of research matters beyond just storm dynamics. Many paleoclimate archives
71 that record past storm activity such as wind-blown dust and sediment, or rainfall linked to westerly
72 winds only capture signals from specific seasons instead of annual averages. If a proxy record
73 mainly reflects one single season, it can either hide or falsely create an orbital signal, depending
74 on which time of year carries the strongest storm response (Laepple et al. 2011). Fully understand-
75 ing which season holds the clearest precessional storm signal is therefore essential for properly
76 interpreting these geological records. Moreover, different metrics for cyclone behavior, e.g., total
77 storm counts, how much rain each storm produces, and storm intensity, react differently to orbital
78 forcing. Researchers must evaluate these quantities separately before drawing conclusions from
79 any single geological proxy.

80 To fill this gap, we run 12 equilibrium climate simulations covering a full precession cycle.
81 Every extratropical cyclone in both hemispheres is tracked individually using 6-hour sea level
82 pressure. Our experimental design isolates precession at amplified eccentricity with all other
83 boundary conditions fixed, so that the seasonal storm response to precession forcing alone can be

84 examined. Section 2 describes our model and simulation, and the method we use for storm tracking
 85 and analysis. Section 3 presents the results, and discuss and conclude in section 4.

86 **2. Methodology**

87 *a. Model and experiments*

88 The simulations are performed with the coupled Earth system model AWI-ESM2 (Sidorenko
 89 et al. 2019). The atmosphere component is ECHAM6 at T63 resolution with 47 vertical levels
 90 (Stevens et al. 2013), which includes the land surface module JSBACH (Raddatz et al. 2007). The
 91 ocean and sea ice component is FESOM2, a finite-volume model formulated on an unstructured
 92 mesh (Danilov et al. 2017), with a spatially varying resolution of from 25 km over high latitudes
 93 and coastal regions, to 150 km over open oceans.

94 12 equilibrium simulations are analyzed, taken from the idealized precession ensemble introduced
 95 by Shi et al. (2025). The longitude of perihelion ω is prescribed at values from 0° to 330° in steps
 96 of 30° , while eccentricity is fixed at 0.058 and obliquity at 24.5° . The eccentricity value sits
 97 near the upper bound reached during the Quaternary and amplifies the precessional modulation of
 98 insolation. All remaining boundary conditions follow the pre-industrial setup, and each phase is
 99 branched from a long pre-industrial control. 30 years of 6-hourly output from the equilibrated part
 100 of each simulation form the basis of the cyclone analysis. The calendar month of perihelion for
 101 each phase is computed from orbital geometry (Berger 1978), and the same formulary provides the
 102 insolation fields shown in Fig. 1.

103 *b. Cyclone detection*

104 Extratropical cyclones are detected and tracked with the algorithm of Crawford et al. (2021),
 105 and Fig. 2 summarizes the procedure with one 6-hourly field of the ensemble. 6-hourly sea level
 106 pressure is first interpolated onto 100-km equal-area grids centered on each pole, and the two
 107 hemispheres are processed separately. A grid cell qualifies as a cyclone center when its pressure
 108 p_c lies below that of all 8 neighbors and the surrounding field keeps a sufficient inward gradient.
 109 The gradient test compares p_c with the mean pressure $\bar{p}(d)$ on a one-cell ring at distance d from
 110 the candidate,

$$\frac{\bar{p}(d) - p_c}{d} \geq \gamma \quad (1)$$

111 where $d = 1000$ km and $\gamma = 750 \text{ Pa (1000 km)}^{-1}$, and candidates over terrain above 1500 m are
 112 discarded. Centers closer than 1000 km are treated as one multicenter system. The cyclone area A
 113 is obtained by growing closed isobars around each center at a 200-Pa interval until a contour fails
 114 to close, and the last closed isobar defines the edge pressure p_e (Fig. 2b). Each cyclone carries the
 115 depth and the area-equivalent radius

$$D = p_e - p_c \quad (2)$$

$$R = \sqrt{A/\pi} \quad (3)$$

116 Tracking links the centers of consecutive 6-hourly fields (Fig. 2c). For every active cyclone a
 117 first-guess position is projected from its previous displacement, damped to three quarters,

$$\mathbf{x}_q = \mathbf{x}_t + 0.75 (\mathbf{x}_t - \mathbf{x}_{t-1}) \quad (4)$$

118 and a center of the next field qualifies as the continuation when it lies within $v_{\max}\Delta t$ of both the
 119 current position $\mathbf{x}_t$ and the projection $\mathbf{x}_q$, with $v_{\max} = 150 \text{ km h}^{-1}$ and $\Delta t = 6 \text{ h}$. Among the
 120 qualifying candidates the one nearest to the projection continues the track. Centers left unmatched
 121 at the new time start tracks as genesis events, and tracks left unmatched end as lysis. Only systems
 122 poleward of 30° latitude enter the analysis.

123 Several track-based diagnostics are derived. Track density counts every 6-hourly cyclone position,
 124 system density counts each cyclone once at its mean position, and genesis density counts the first
 125 point of each track. Tracks that merely cross the boundaries of the monthly processing windows
 126 would appear as spurious genesis or lysis events, and such truncated events (about 7% of the
 127 total) are removed. For every cyclone the lifespan, the maximum depth relative to the outermost
 128 closed isobar, the minimum central pressure, the mean cyclone area, and the propagation speed are
 129 recorded.

130 *c. Gridded storm coverage*

131 A complementary Eulerian diagnostic measures how long each location spends under cyclone
 132 influence. At every six-hourly time step a binary mask flags the grid cells that lie inside a square
 133 search window of half-width $\sqrt{A}$ grid cells around each cyclone center and whose pressure lies

below the edge pressure of that cyclone,

$$M(\mathbf{x}, t) = 1 \quad (5)$$

where $p(\mathbf{x}, t) < p_e$ inside the window, and the masks are regridded to T63 and accumulated into monthly storm days, one quarter of the flagged six-hourly steps. Storm days at a point therefore scale with the product of cyclone number and cyclone area footprint. The two factors can carry opposite seasonal cycles, and over the Southern Ocean the larger size of summer cyclones raises summertime storm days even though cyclone counts peak in winter. Frequency statements in this paper consequently rest on the track-based densities, while storm days serve as a smooth field for maps and for measures of cyclone influence time. The gridded product covers all twelve phases. An early processing inconsistency in the $\omega = 30^\circ$ mask product was traced to a stale intermediate file and removed by regenerating the storm masks of all twelve experiments with one pipeline version, and the regenerated fields were verified month by month against the Lagrangian statistics.

An Eulerian measure of synoptic variability provides a further check that involves no tracking at all. Bandpass sea level pressure variability is computed from the 6-hourly fields with the 24-h difference filter of Wallace et al. (1988), which isolates fluctuations on the synoptic time scales that define the classical storm tracks (Blackmon 1976), and its monthly standard deviation is analyzed with the same basins, seasons, and response measures.

d. Baroclinicity diagnostics

The maximum Eady growth rate serves as the measure of baroclinicity (Eady 1949; Lindzen and Farrell 1980; Hoskins and Valdes 1990). It is computed as $\sigma = 0.31 |f| |\partial U / \partial z| N^{-1}$ in the 925–700 hPa layer from monthly wind, temperature, and geopotential height. The response to precession is quantified between the end members p090 and p270, which place perihelion at the June and December solstices. Because the difference fields are dipolar, plain box averages cancel, and the response amplitude is instead defined as the root-mean-square of the difference over a basin, normalized by the climatological growth rate of the season. A substitution decomposition in the style of Camargo et al. (2007) splits the difference field into a vertical-shear factor and a static-stability factor, and the share of each factor is obtained by projecting it onto the full difference pattern. The basin averages use the storm-track boxes defined in the next sub-section. beginning

at 30°S would admit the seasonal signal of the subtropical winter jet and disguise it as a storm-belt response.

e. Basins and statistics

Three storm-track basins are analyzed, the North Atlantic (30–75°N, 80°W–20°E), the North Pacific (30–75°N, 120°E–120°W), and the Southern Ocean (40–75°S, all longitudes). Basin means are area weighted. The size of the precessional response is reported as the range across the twelve phases, the maximum minus the minimum of the 30-year mean, expressed as a percentage of the phase mean. Uncertainty estimates use the interannual spread of the 30 individual years in each phase, and the significance of the end-member difference maps is assessed with a two-sided Welch t test at each grid point. exceed the interannual noise, with p values below 10^{-12} in all three basins.

3. Results

a. The summer paradox

The 12 precession-phase annual midlatitude storm climatologies exhibit only minor differences, and phase anomalies relative to the multi-phase ensemble mean manifest as weak, slowly evolving dipoles aligned along major storm track pathways (Fig. 3). The basin-averaged time series quantify the muted amplitude of this annual-mean signal, with across-phase ranges of 10%, 17% and 4% in the North Atlantic, the North Pacific and the Southern Ocean, varying smoothly with ω . We also observe an interhemispheric antiphase.

Beneath this annual background lies a pronounced seasonal cycle of storm activity. Extratropical cyclone activities peak in DJF in Northern Hemisphere basins and JJA across the Southern Ocean, with this seasonal timing unchanged across all orbital setups. Cold-season cyclones also carry greater intensity. Winter peak cyclone depths exceed summertime values by approximately 50 % within the Northern Hemisphere storm basins (Fig. 4a–c; Table 1). Overall, winter dominates midlatitude cyclone activity uniformly across hemispheres and all orbital phases.

Precession exerts negligible control over annual-mean storm conditions but profoundly reshapes seasonal cyclone variability, and its strongest modulation targets the low-activity warm season. Though each precession phase retains the same wintertime peak in basin-averaged storm track density, cross-phase deviations emerge almost entirely within each basin’s local summer (Fig. 4a–c).

189 Seasonal range statistics quantify this stark seasonal contrast in orbital sensitivity (Fig. 4d–f). For
190 the North Pacific, the total across-precession variability accounts for just 7 % of the phase mean on
191 an annual basis, yet rises to 44 % in boreal summer (JJA). The North Atlantic follows an identical
192 pattern, with annual variability of 7 % versus a summertime range of 25 %. The Southern Ocean
193 mirrors this behavior but shifts the sensitive season to its local summer (DJF), with an annual range
194 of only 4 % compared to a 22 % summer signal. All basins uniformly identify local summer as
195 the orbital-sensitive season, and the small annual signal is what survives of a substantial summer
196 swing after the stable winter is averaged in.

197 This summertime storm response is locked to orbital geometry. When basin-averaged track-
198 density anomalies are plotted against both calendar month and precession angle ω , the band of
199 peak anomalies traces a diagonal ridge aligned with the calendar timing of perihelion across
200 the full phase space (Fig. 4g–i), with a delay of 1–2 months. Summers spent near perihelion
201 exhibit suppressed storm activity, whereas aphelion summers feature intensified storminess. A
202 single precessional forcing thus drives opposing midlatitude storm responses between the two
203 hemispheres across the orbital cycle, as northern summer perihelion aligns with southern summer
204 aphelion under orbital precession, and vice versa, reversing the insolation control on warm-season
205 storminess between hemispheres.

206 *b. The dynamical bridge runs through summer shear*

207 The pronounced seasonal dependence of orbital storm signals originates directly from insolation
208 forcing. Averaged across the Northern Hemisphere storm-track band (40–70°N), precessional
209 insolation variability peaks at 105 W m^{-2} during boreal summer (JJA) and falls to merely 14 W m^{-2}
210 in boreal winter (DJF). The Southern Hemisphere storm band (40–70°S) exhibits mirror-symmetric
211 behavior, with a maximum seasonal forcing amplitude of 119 W m^{-2} in austral winter (DJF) versus
212 just 47 W m^{-2} in austral summer (JJA) (Fig. 1). The underlying mechanism lies in the multiplicative
213 nature of precessional insolation modulation. Shifting perihelion across the annual cycle alters
214 the Sun–Earth distance for each month, which rescales monthly incoming solar flux by a factor
215 proportional to eccentricity e (approximately $4e$, equaling 23 % for $e = 0.058$). Midwinter at
216 midlatitudes features minimal climatological insolation, leaving little baseline flux to be modulated

by precessional shifts. Precession forcing therefore exerts its strongest modulation on local summer in each hemisphere, and barely perturbs deep winter conditions.

Orbital insolation anomalies are translated into extratropical storm activity via shifts in midlatitude baroclinicity. Differences in maximum Eady growth rate between the extreme precession phases $\omega = 90^\circ$ and $\omega = 270^\circ$ reach their peak magnitude within each basin's local summer (Fig. 5). Normalized against seasonal climatological means, summertime Eady growth rate responses hit 31 % in the North Atlantic, a pronounced 92 % in the North Pacific, and 21 % in the Southern Ocean. A decomposition analysis further reveals that vertical wind shear dominates the precessional Eady growth rate anomalies, while static stability contributes negligible signal strength. This is consistent across all storm tracks. Perihelion summers warm the summer hemisphere, flatten the meridional temperature gradient, weaken the thermal-wind shear, and suppress the baroclinic growth that sustains warm-season extratropical cyclones. The storm anomalies appear somewhat poleward of the growth-rate anomalies, consistent with the downstream development of baroclinic waves (Chang et al. 2002).

c. Spatial expression, and what precession actually moves

Spatial difference maps between the two orbital end-members (p090 minus p270) yields weak responses for the annual means (Fig. 6a). By contrast, boreal summer (JJA) exhibits a prominent dipole pattern across Northern Hemisphere basins, with fewer storm days through the midlatitude core and more along the Arctic flank at perihelion summer, the signature of a poleward shift joined to an overall summer weakening. In DJF the signal migrates to the Southern Ocean and forms a circumpolar banded structure of opposite sign, marking enhanced midlatitude storm activity and an equatorward shift of the southern storm belt.

The latitudinal displacement of storm tracks driving these dipole anomalies is quantified in Fig. 7. For the North Pacific boreal summer, core storm latitude shifts from 53°N when perihelion falls in northern winter to 61°N when it falls in northern summer, yielding an 8° total poleward migration that peaks precisely at $\omega = 90^\circ$. The North Atlantic summertime storm track undergoes a 4.4° poleward shift following identical orbital behavior, while the Southern Ocean mirrors this response with a 3.8° poleward displacement when perihelion coincides with austral summer (near p270). A perihelion summer therefore weakens storm activity and shifts the storm population poleward,

246 consistent with the poleward shift of baroclinic activity evident in the Eady growth rate dipole
247 (Fig. 5a).

248 By contrast, the annual-mean track latitude varies by no more than 1.3° in any basin. Winter
249 storm-track latitudes only fluctuate by $1.5\text{--}2.8^\circ$ across the full precession cycle, and their latitudinal
250 extrema do not align with solstitial orbital phases. This further confirms our earlier conclusion
251 that midlatitude winter storms are orbitally insensitive. Spring and autumn display intermediate
252 latitudinal variability with a coherent phase dependence. For example, autumn poleward displace-
253 ment maximizes near $\omega = 150^\circ$, the orbital configuration that places perihelion in late summer.
254 Latitudinal poleward shifts therefore track the calendar timing of perihelion identically to Fig. 4g–i,
255 with a uniform lag of 1–2 months driven by surface thermal inertia. Figure 8 quantifies this basin-
256 dependent lag. All basin-scale storm extrema cluster along basin-specific dashed lines offset from
257 the perihelion reference. The response lags the perihelion calendar by distinct intervals across
258 regions, near one month in the North Pacific, 1–2 months in the North Atlantic, and slightly over
259 two months in the Southern Ocean.

260 *d. What the rain records*

261 The hydrological signature of precessional forcing evolves in antiphase with storm dynamical
262 responses (Fig. 9). Detected extratropical cyclones contribute roughly one-third of total midlatitude
263 precipitation across our simulations, consistent with storm precipitation partitioning derived from
264 observational reanalysis datasets (Hawcroft et al. 2012). During boreal summer over the North
265 Pacific, mean precipitation per individual storm rises by as much as 15 % at the orbital phase where
266 total storm frequency reaches its minimum, yielding nearly perfectly opposing variability between
267 the two metrics. Perihelion summers feature warmer, more humid midlatitude atmospheres, which
268 amplifies precipitation intensity within the remaining cyclones; nevertheless, the sharp reduction
269 in storm number dominates the total signal, such that basin-wide summertime precipitation drops
270 by 17 % between the two extreme orbital phases. The fractional share of precipitation originating
271 from cyclones also declines markedly, with a full-cycle range of 9.5 % in North Pacific summer
272 compared to just 3 % for the annual mean. Precessional forcing thus exerts two counteracting
273 controls on extratropical hydroclimate simultaneously. It reduces total cyclone counts while
274 boosting precipitation intensity per storm.

4. Discussion

The seasonal preference seen in storm responses originates purely from orbital geometry, independent of model physics. Precession redistributes insolation in proportion to the insolation a season already has, meaning winter hemispheres have very little background insolation for orbital changes to modify (Berger 1978). Fig. 1 clearly illustrates this stark seasonal contrast in forcing strength. Winter insolation varies by only roughly 20 W m^{-2} across the precession cycle, while summer values exceed 125 W m^{-2} within mid-latitude storm-track latitudes. This is model independent and will hold in any climate models as it simply reflects the orbital geometry behind the well-known summer-dominated orbital forcing framework (Huybers 2006). Existing monsoon studies has long confirmed that summer bears the strongest precessional signal (Kutzbach and Guetter 1986; Bosmans et al. 2015). But the new insight of our work lies elsewhere. This summer-focused orbital forcing acts on a winter-dominated weather system, reshaping storm activity exclusively during its normally quiet warm season. Note that the term "quiet" here refers to the weaker overall intensity of the summer storm track, not its population. Both hemisphere still hosts abundant storm systems in local summer, and they are merely weaker, shallower and move more slowly compared to winter cyclones.

The mechanism that converts summer insolation into summer storms is the thermal wind. When perihelion falls in local summer, the corresponding hemisphere experiences a warming over high latitudes. This flattens the meridional temperature gradient and reduces vertical wind shear that drive baroclinic growth. Our factor decomposition analysis confirms that the shear term carries essentially the whole Eady growth rate response in all three ocean basins, while static stability plays a negligible role. We see similar physical mechanism operating in the modern Northern Hemisphere, where greenhouse warming takes the place of perihelion-driven heating. Stronger summer warming at high latitudes has weakened summertime large-scale circulation, eddy kinetic energy, and the number of strong summer cyclones (Coumou et al. 2015; Chang et al. 2016; Gertler and O’Gorman 2019). Future climate projections from standard CMIP models further show that summer cyclone activity and baroclinic energy will decline via this same shear-based mechanism as global warms (Lehmann et al. 2014; Priestley and Catto 2022). But a difference should be noted here, as long-term greenhouse warming warms both hemispheres simultaneously, whereas

precession swings them in antiphase. Even so, both forcings follow one unified rule that is the core theme of this study: cooler summers host more storm activity.

Compared to summer, storm systems in winter are much more stable. Across all three ocean basins, winter storm activity changes by less than 10% over a full precession cycle, despite that winter Eady growth rates shift by 16–18% between the two orbital end-members, and weak winter insolation forcing cannot fully explain this stability. The winter storm track appears saturated in the sense that ample baroclinicity is available in every phase, so that moderate changes in the growth rate do not translate into changes in occurrence. A transient simulation of the last millennium likewise found cyclone counts nearly insensitive to external forcing (Raible et al. 2018; Hess and Chemke 2025). Our findings expand this conclusion to the far larger swings in solar radiation driven by precession, and this insensitivity only holds for wintertime conditions.

Thus, Geological records sensitive to summertime conditions within midlatitude storm belts should preserve prominent precession-scale storm signals, whereas proxies that average climate signals across winter or the full year will show only minor orbital imprint. Just as observed for temperature reconstructions, the seasonal bias of a given archive can either generate artificial orbital storm signals or fully obscure true precessional variations (Laepple et al. 2011). Furthermore, summer-sensitive storm records from both hemispheres will fluctuate oppositely over precessional cycles, forming an extratropical equivalent of the well-documented monsoon seesaw (Wang et al. 2008). Current published sediment records already support the feasibility of testing this framework. Distinct precession-period signals have been detected in Northern Hemisphere midlatitude westerly proxies (Li et al. 2019), as well as North Pacific dust records. Precessional variation appears specifically in moisture-sensitive dust fractions, while the annually averaged dust flux responds primarily to obliquity forcing (Zhong et al. 2024). Meanwhile, dust intensity archives that capture winter-only signals display very weak precessional power (Chen et al. 2014). Regarding the Southern Hemisphere, subtropical margins of the southern westerly belt exhibit clear precession cycles (Lamy et al. 2019), consistent with our results. And the presence or absence of precession signals at individual sites depends on their latitudinal position within the wind belt (Kaiser et al. 2024). Records of the southern westerlies also show orbital-band migrations of the belt itself (Lamy et al. 2010; Varma et al. 2012), and glacial-state cyclone modeling has shown how strongly

reconstructed storminess depends on which cyclone property a proxy senses (Pinto and Ludwig 2020).

Although the three storm basins follow the same physical mechanism, they exhibit different regional behaviors, which can be distinguished through separate diagnostic metrics. In local summer, storm frequency and track position vary substantially by 16–68% across the precession cycle, while storm propagation speed remains largely unchanged. By contrast, storm intensity exhibits clear basin-dependent differences. The Southern Ocean shows a 22% swing in storm frequency but only a 4% change in storm intensity. The North Pacific features a systematic reduction in overall storm activity during perihelion summers. This is consistent with previous findings from anthropogenic warming scenarios that storm frequency and intensity respond to external forcing through independent physical mechanism (Bengtsson et al. 2009; Ulbrich et al. 2009; Catto et al. 2019). Hydrological responses further amplify this contrast. Individual storms produce stronger precipitation in summers with reduced cyclone numbers, indicating that hydrological adjustments exceed dynamical changes under orbital forcing, a pattern analogous to that identified under greenhouse warming (Bengtsson et al. 2009). Given that extratropical precipitation is predominantly generated by cyclones and frontal systems (Hawcroft et al. 2012; Catto and Pfahl 2013), this competing effect substantially alters precipitation partitioning. So a precipitation proxy tend to record combined signal of reduced storm occurrence and intensified single-storm rainfall during perihelion summers.

Our study is based on idealized simulations designed to isolate pure precessional forcing. The experimental setup holds eccentricity, obliquity, ice sheets, and greenhouse gas concentrations fixed and does not incorporate transient climate evolution or orbital cross-interactions. As such, our results represent dynamical responses to idealized orbital conditions rather than reconstructions of specific geological epochs. Future work could incorporate time-varying boundary conditions and full orbital combinations to further generalize the storm seasonal response across broader paleoclimate backgrounds.

5. Conclusions

In our study, we carry out a set of equilibrium climate simulations covering a full precession cycle, with explicit tracking of all extratropical cyclones across both hemispheres to isolate precessional

362 forcing. We find that precession regulates extratropical cyclone primarily through local summer,
363 the season with naturally low storm activity, while winter storm tracks has only minor change
364 across all precession phases. Storm activity calms down when perihelion falls in local summer
365 and intensifies when aphelion coincides with local summer. And, as perihelion summer in one
366 hemisphere matches aphelion summer in the other, mid-latitude storminess of the two hemispheres
367 varies in antiphase during a precession cycle. This orbital signal propagates through shear-driven
368 baroclinicity and expresses itself in cyclones numbers and position, both of which are phase locked
369 to the calendar month of perihelion, with a delay of 1-2 months. Our result also bring clear
370 implications for paleoclimate proxy interpretation. Orbital storm signals should be extracted from
371 midlatitude archives sensitive to summer climate, and the recorded signals will display opposite
372 variations between the two hemispheres.

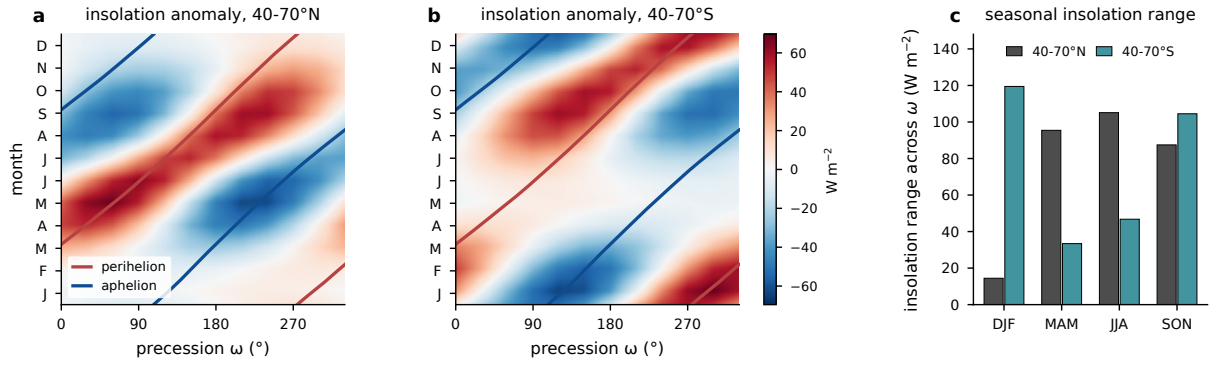


FIG. 1. Precessional insolation forcing from orbital geometry with eccentricity 0.058 and obliquity 24.5° . (a) Monthly insolation anomaly relative to the phase mean, averaged over the storm-track band $40\text{--}70^\circ\text{N}$, as a function of calendar month and precession phase ω , with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid. (b) As in (a), but for $40\text{--}70^\circ\text{S}$. (c) Seasonal forcing amplitude in the two bands, the seasonal mean of the monthly maximum minus minimum insolation across the twelve phases. Units: W m^{-2} .

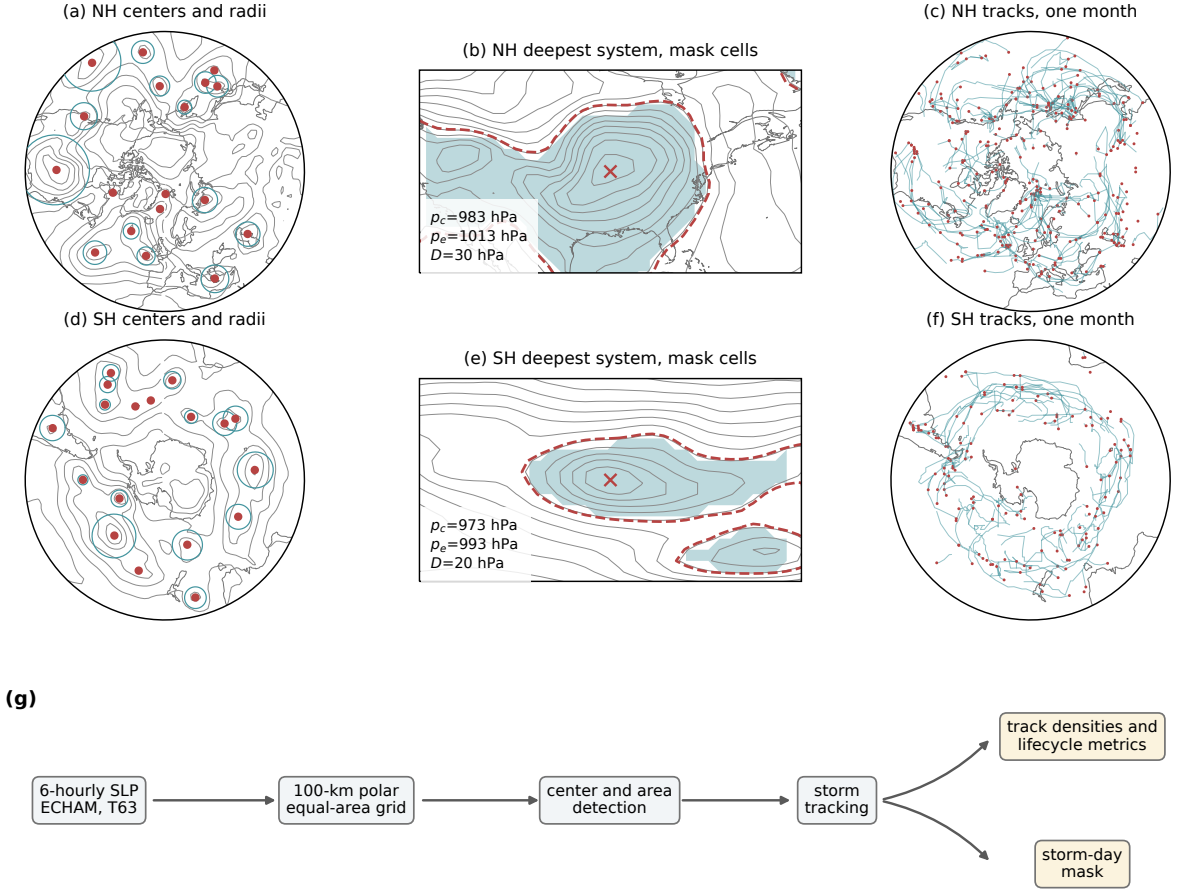


FIG. 2. Cyclone detection and tracking procedure. (a,c) One 6-hourly sea level pressure field of phase p000 (contours, 8-hPa interval) with all detected cyclone centers poleward of 30°N (dots) and their area-equivalent radii $R = \sqrt{A/\pi}$ (circles), 15 January of one model year. (b,e) The deepest system of that field, with the outermost closed isobar p_e (dashed), the cyclone center (cross), the central and edge pressures and depth annotated, and the binary mask cells (shading) defined by $\text{SLP} < p_e$ inside the $\sqrt{A}$ search window; contour interval 4 hPa. (c,f) All cyclone tracks of the same month (lines) with genesis points (dots). (d) Data processing workflow.

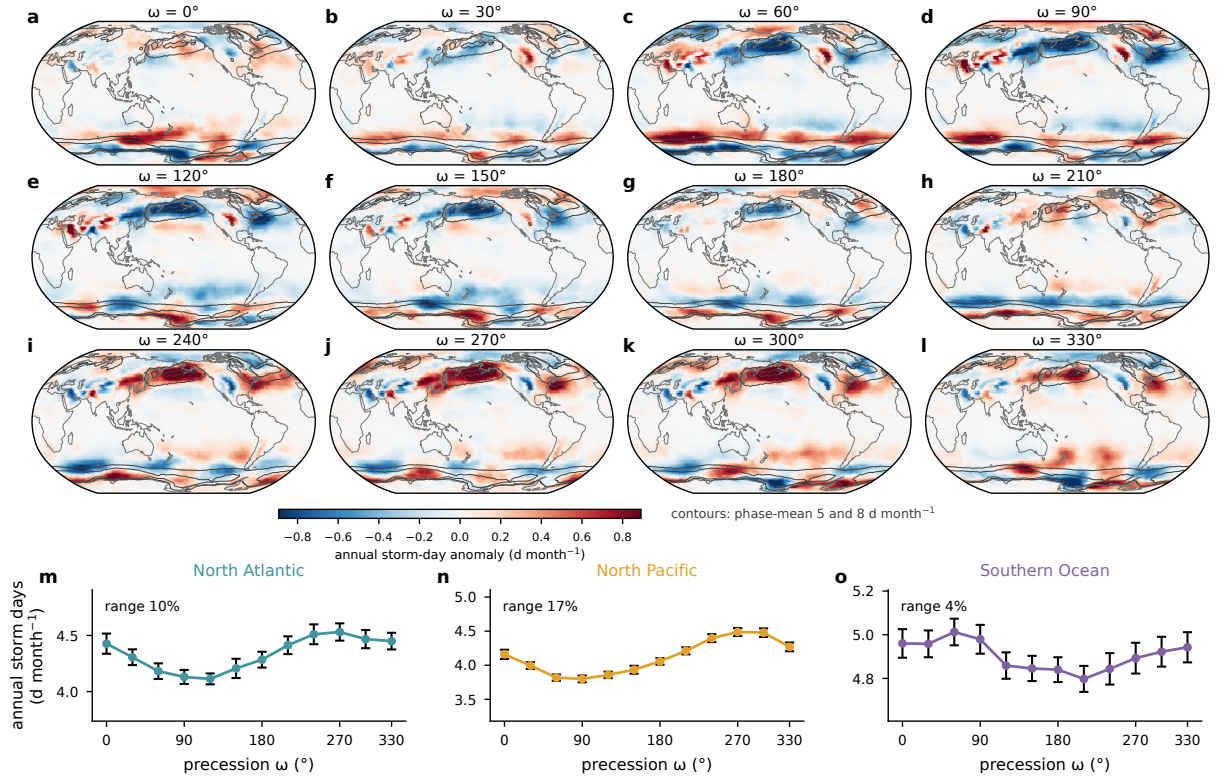


FIG. 3. Annual-mean storm days across the precession cycle. (a)–(l) Annual storm-day anomaly of each phase relative to the twelve-phase mean (shading), with the phase-mean climatology outlined at 5 and 8 d month⁻¹ (contours). (m)–(o) Basin-mean annual storm days against ω with interannual standard errors; the across-phase range as a percentage of the phase mean is annotated in each panel.

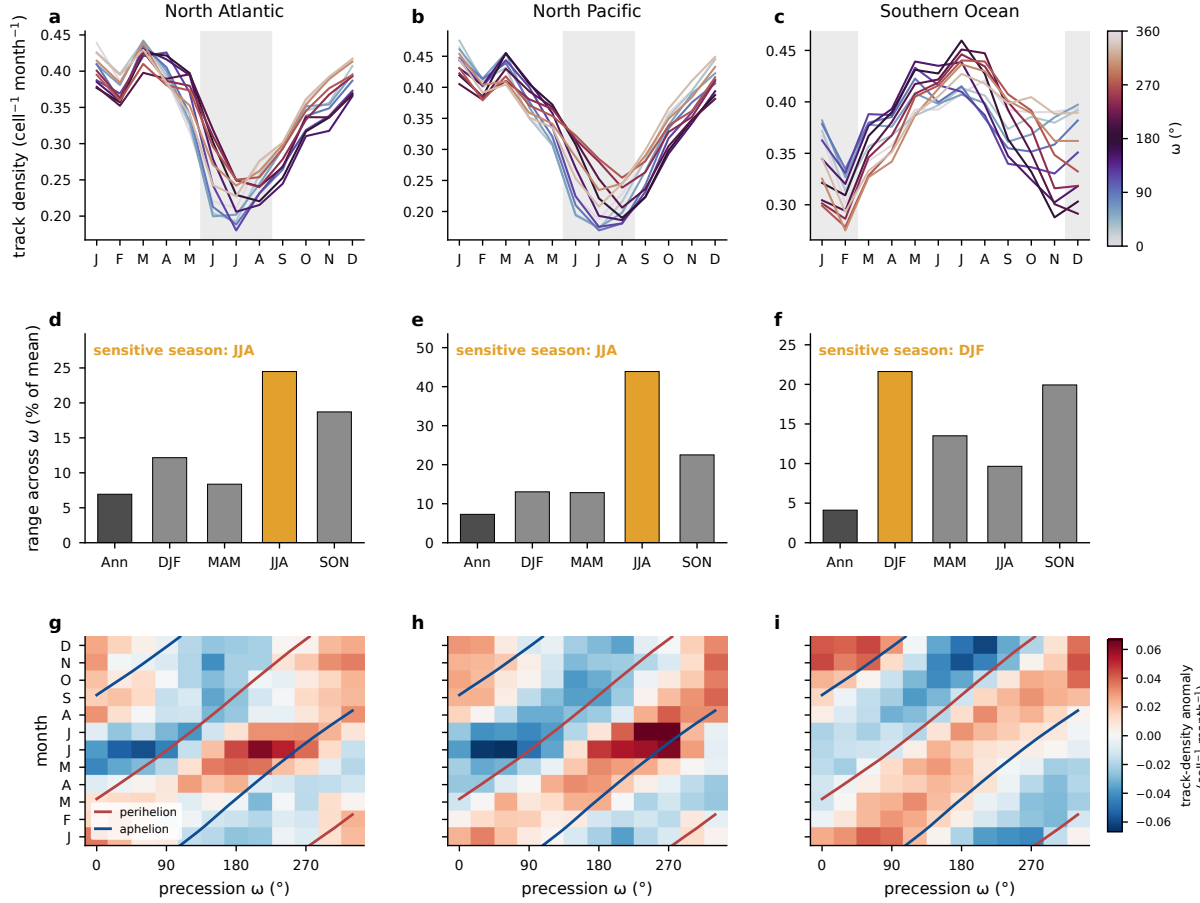


FIG. 4. Seasonal concentration of the precessional storm response, from the Lagrangian tracks. (a)–(c) Basin-mean track density (cell⁻¹ month⁻¹) by calendar month for the North Atlantic, the North Pacific, and the Southern Ocean, one line per phase colored by ω , with local summer shaded. (d)–(f) Across-phase range of track density (maximum minus minimum, percent of the phase mean) for the annual mean and the four seasons, with the local-summer bar highlighted; open symbols give the corresponding ranges of system and genesis densities. (g)–(i) Basin-mean track-density anomaly relative to the phase mean as a function of calendar month and precession phase (unsmoothed), with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid.

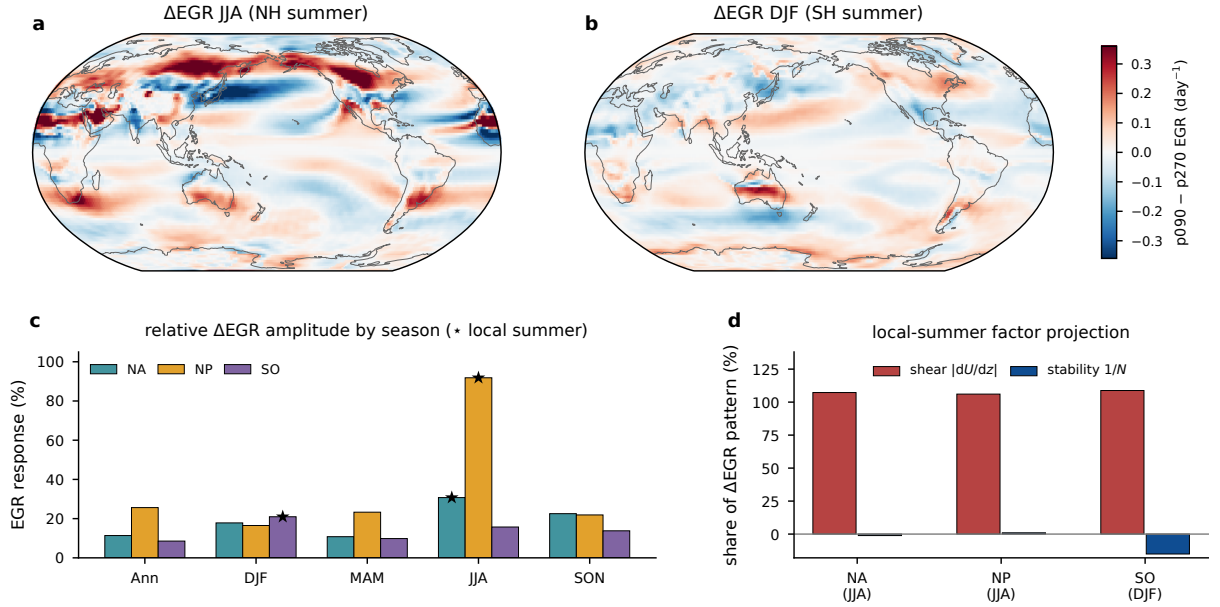


FIG. 5. Precessional response of the maximum Eady growth rate (925–700 hPa). (a) p090 minus p270 difference for JJA and (b) for DJF (day^{-1}). (c) Seasonal response amplitude in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology, with stars marking local summer. (d) Projection shares of the vertical-shear and static-stability factors in the local-summer difference pattern.

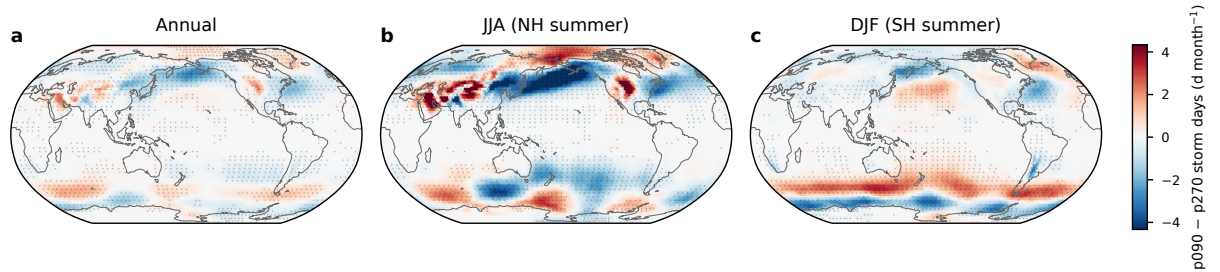


FIG. 6. Difference in storm days (d month^{-1}) between p090 and p270 for (a) the annual mean, (b) JJA, and (c) DJF, on a common color scale. Stippling marks grid points where the difference is significant at the 95% level based on a two-sided Welch t test over the 30 simulated years.

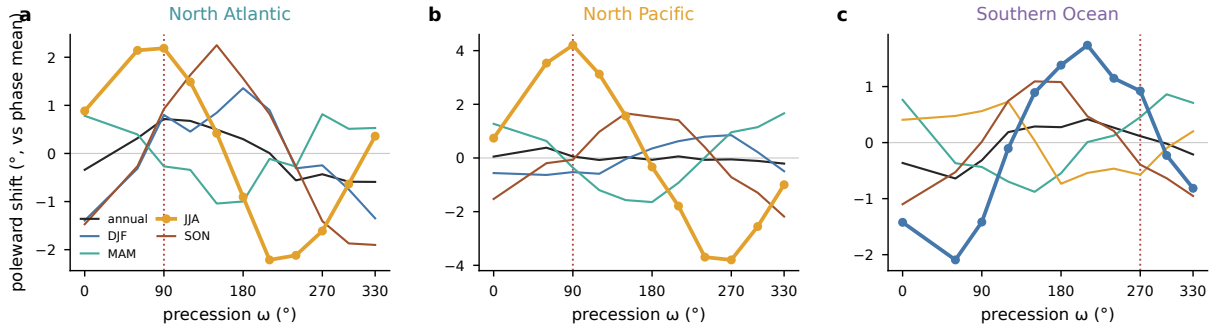


FIG. 7. Storm-track latitude across the precession cycle, the storm-day-weighted mean latitude over each basin sector shown as the poleward anomaly from each season's phase mean, for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean. Lines give the annual mean and the four seasons, with the local summer of each basin drawn heavy with markers. Vertical dotted lines mark the phase with perihelion in the local summer of each basin.

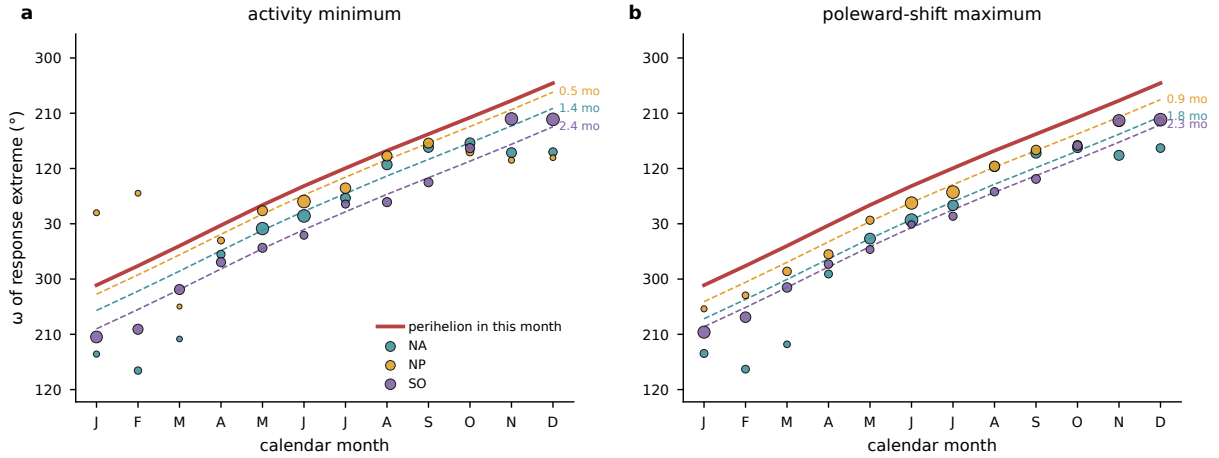


FIG. 8. Phase locking of the storm response to the perihelion calendar. For each calendar month and basin, the phase of the first harmonic across the 12 precession phases of (a) basin-mean storm days, shown as the ω of the activity minimum, and (b) the storm-day-weighted track latitude, shown as the ω of the poleward maximum. Marker size scales with the harmonic amplitude. The red line gives the ω that places perihelion in that calendar month, and dashed lines repeat it delayed by each basin's median lag over the amplitude-strong months (annotated in months).

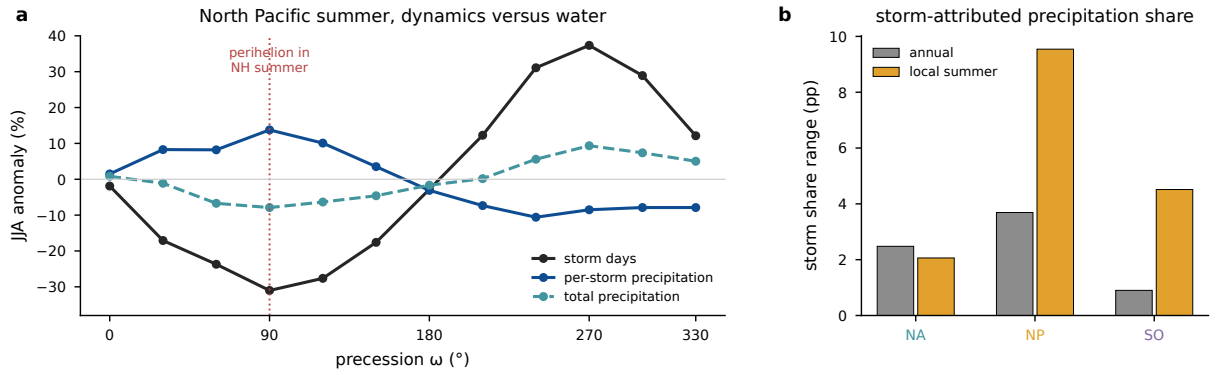


FIG. 9. Storm-attributed precipitation across the precession cycle. (a) North Pacific JJA percent anomalies from the phase mean of basin-mean storm days, per-storm precipitation, and total precipitation; the dotted line marks the phase with perihelion in NH summer. (b) Across-phase range of the storm-attributed share of precipitation, accumulated precipitation inside storm footprints divided by total accumulated precipitation (percentage points), for the annual mean and local summer in each basin.

TABLE 1. Three-basin comparison of the precessional storm response. Ranges are max minus min across the twelve phases as a percentage of the phase mean. Storm and sensitive seasons refer to local winter and local summer. The storm season is diagnosed from cyclogenesis counts, the sensitive season from the seasonal storm-day range. The EGR response is the RMS p090–p270 difference over the basin sector relative to its climatology, and the shear share is the projection of the vertical-shear factor onto the response pattern.

Basin	Storm season	Sensitive season	Storm-day range ann/summer (%)	Summer max/min ω (°)	EGR response summer (%)	Shear share (%)
North Atlantic	DJF	JJA	10 / 32	240 / 90	31	107
North Pacific	DJF	JJA	17 / 68	270 / 90	92	106
Southern Ocean	JJA	DJF	4 / 16	60 / 210	21	109

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477 *Data availability statement.* The AWI-ESM simulations of the idealized precession ensemble
478 are described in Shi et al. (2025). The cyclone-track database, the gridded storm statistics, and
479 the analysis scripts underlying all figures will be archived and assigned a DOI upon acceptance.
480 [TODO repository DOI.]

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